REPORT LAB 6:

DETERMINING UNKNOWN RESISTANCE AND CAPACITANCE USING NEON LAMP OSCILLATOR CIRCUIT

Class: CC10 / Group: 1		Lecturer's comment
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I. Aims/Purposes

- To study the accumulate-released oscillation circuit using the neon light by testing the equations

- To find the value of an unknown resistance R_x and an unknown capacitance C_x arising in an accumulate-released oscillation circuit.

II. Apparatus, Methods, and Procedure

1) <u>Apparatus</u>

- The physics experiment equipment MC – 958
- The multifunction versatile chronometer MC-963A
- A sample resistor with resistance $R_0 = 1.00 \pm 0.01$ MΩ
- A resistor of unknown resistance R_x
- A protective resistor with R = 100kΩ
- A sample capacitor with capacitance $C_0 = 1.00 \pm 0.01$ μF
- A capacitor of unknown capacitance C_x
- An infrared photoelectric sensor.

2) <u>Methods and procedures</u>

a) Measure U_b (the on electric potential) and U_e (the off electric potential) of the neon light

- Firstly, we connect a resistor R = 100kΩ in series with the neon light and a voltmeter parallel to the neon light. Then we turn on the switch of the electric supply U_n to increase electric potential U by spinning the knob until the neon light glows. We can now measure U_b (equals to U at this moment). Finally, we turn back the knob of the supply U_n to decrease the potential U between the two poles of neon light Ne until the light go out. We can now measure U_e (equals to U at this moment). We repeat this experiment 5 times to get the average value.

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b) Test the formula to determine period τ of the accumulate-released oscillation circuit

- Firstly, we construct the circuit same as previous. On the MC-963A, we turn the knob so that n = 50. Brushing the time scale selector knob "TIME RANGE" to position 99.99s. Next, we close the circuit, set the voltmeter to U = 90V and keep this value constant.

- After that, we hook the chronometer with a 220V power supply and press "RESET". The chronometer will now automatically measure the interval time t_0 (n = 50) so we write down this value and repeat this experiment 5 times. Finally, dividing t_0 by n = 50, we obtain τ_0.

c) Determine the resistance R_x

- Swap the sample resistor R_0 with R_x and turn on the switch K. Turn the knob of the supply U_n until voltmeter V reaches U_n = 90V and keep this value constant during the experiment and press "RESET". The chronometer will now automatically measure the interval time t_x of n = 50 of the period τ_x so we write this value t_x down and repeat the experiment 5 times. Next, dividing t_x by n = 50, we obtain τ_x and applying the overall equation allows us to determine R_x.

d) Determine the capacitance C_x

- Swap the sample capacitor C_0 with C_x and turn on the switch K. Turn the knob of the supply U_n until voltmeter V reaches U_n = 90V and keep this value constant during the experiment and press "RESET". The chronometer will now automatically measure the interval time t'_x of n = 50 of the period τ'_x so we write this value t'_x down and repeat the experiment 5 times. Next, dividing t'_x by n = 50, we obtain τ'_x and applying the overall equation allows us to determine C_x.

III. Equations

Calculate the period τ of the accumulate-released oscillation circuit

$$\tau \approx R_0 . C_0 . ln \frac{U_n - U_e}{U_n - U_b}$$

IV. Experimental data

Table 1

- Voltmeter V: $U_m = 100V$; $\delta_v = 2.5\%$
- Accuracy of the chronometer: $\Delta t_{sys} = 0.01\ sec$
- $C_0 = 1.00 + 0.01\ (\mu F)$
- $R_0 = 1.00 + 0.01\ (M\Omega)$

	U_b (V)	ΔU_b (V)	U_e (V)	ΔU_e (V)	t_0 (sec)	Δt_0 (sec)	t_R (sec)	Δt_R (sec)	t_C (sec)	Δt_C (sec)
1	72.0	1.60	62.0	0.80	38.83	0.020	122.79	1.110	22.00	0.030
2	74.0	0.40	62.0	0.80	38.81	0.040	122.76	1.080	21.97	0.000
3	74.0	0.40	60.0	1.20	39.02	0.170	121.04	0.640	21.94	0.030
4	74.0	0.40	60.0	1.20	38.97	0.120	120.89	0.790	21.97	0.000

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5	74.0	0.40	62.0	0.80	38.62	0.230	120.92	0.760	21.97	0.000
Avg.	73.60	0.64	61.20	0.96	38.850	0.116	121.680	0.876	21.970	0.012

V. Calculation works

Calculating the absolute error

$$\Delta U_{s_{sys}} = U_m \,.\, \delta_v = 100 \times 2.5\% = 2.50\ (V)$$

$$\Delta U_n = \Delta U_{n_{sys}} + \Delta \overline{U_n} = 2.5 + 0 = 2.50\ (V)$$

$$\Delta U_b = \Delta U_{b_{sys}} + \Delta \overline{U_b} = 2.5 + 0.64 = 3.14\ (V)$$

$$\Delta U_e = \Delta U_{e_{sys}} + \Delta \overline{U_e} = 2.5 + 0.96 = 3.46\ (V)$$

$$\Delta t_0 = \Delta t_{0_{sys}} + \Delta \overline{t_0} = 0.01 + 0.116 = 0.126\ (s)$$

$$\Delta t_R = \Delta t_{R_{sys}} + \Delta \overline{t_R} = 0.01 + 0.876 = 0.886\ (s)$$

$$\Delta t_C = \Delta t_{C_{sys}} + \Delta \overline{t_C} = 0.01 + 0.012 = 0.022\ (s)$$

Calculating the value of quantities

- Indirect measurement:

$$\tau_0 = R_0 C_0 ln\left(\frac{U_n - U_e}{U_n - U_b}\right) = 1\text{x}10^6 \times 1\text{x}10^{-6} \times \ln\left(\frac{90 - 61.20}{90 - 73.60}\right) = 0.563\ (s)$$

$$\begin{aligned}\Delta\tau_0 = {} & (C_0 \Delta R_0 + R_0 \Delta C_0) ln \frac{U_n - U_e}{U_n - U_b} \\ & + R_0 C_0 \left(\left|\frac{1}{U_n - U_e} - \frac{1}{U_n - U_b}\right| \Delta U_n + \left|\frac{-1}{U_n - U_e}\right| \Delta U_e \right. \\ & \left. + \left|\frac{1}{U_n - U_b}\right| \Delta U_b\right) \\ & = (1\text{x}10^{-6} \times 0.01\text{x}10^6 + 1\text{x}10^6 \times 0.01\text{x}10^{-6}) \ln\left(\frac{90 - 61.20}{90 - 73.60}\right) \\ & + 1\text{x}10^6 \times 1\text{x}10^{-6} \\ & \times \left(\left|\frac{1}{90 - 61.20} - \frac{1}{90 - 73.60}\right| \times 2.50 + \left|\frac{-1}{90 - 61.20}\right| \times 3.46 \right. \\ & \left. + \left|\frac{1}{90 - 73.60}\right| \times 3.14\right) = 0.388\ (s)\end{aligned}$$

- Direct measurement:

$$\tau_0 = \frac{t_0}{n} = \frac{t_0}{50} = \frac{38.85}{50} = 0.777\ (s)$$

$$\Delta\tau_0 = \frac{\Delta t_0}{50} = \frac{0.126}{50} = 0.003\ (s)$$

$$R_x = R_0 \frac{t_R}{t_0} = R_0 \frac{\tau_R}{\tau_0} = 1\text{x}10^6 \times \frac{121.680}{38.850} = 3.132\ (M\Omega)$$

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$$\frac{\Delta R_x}{R_x} = \frac{\Delta R_0}{R_0} + \frac{\Delta t_R}{t_R} + \frac{\Delta t_0}{t_0} = \frac{0.01\text{x}10^6}{1\text{x}10^6} + \frac{0.886}{121.680} + \frac{0.126}{38.850} = 0.021$$

$$\Rightarrow \Delta R_x = 0.021 \times 3.132\text{x}10^6 = 0.066\ (M\Omega)$$

$$C_x = C_0\ \frac{t_C}{t_0} = C_0\ \frac{\tau_C}{\tau_0} = 1\text{x}10^{-6} \times \frac{21.970}{38.850} = 0.566\ (\mu F)$$

$$\frac{\Delta C_x}{C_x} = \frac{\Delta C_0}{C_0} + \frac{\Delta t_C}{t_C} + \frac{\Delta t_0}{t_0} = \frac{0.01\text{x}10^{-6}}{1\text{x}10^{-6}} + \frac{0.022}{21.970} + \frac{0.126}{38.850} = 0.014$$

$$\Rightarrow \Delta C_x = 0.014 \times 0.566\text{x}10^{-6} = 0.008\ (\mu F)$$

Conclusion:

- Indirect measurement:

$$U_n = \overline{U_n} \pm \Delta U_n = 90 \pm 2.50\ (V)$$

$$U_b = \overline{U_b} \pm \Delta U_b = 73.60 \pm 3.14\ (V)$$

$$U_e = \overline{U_e} \pm \Delta U_e = 61.20 \pm 3.46\ (V)$$

$$\tau_0 = \overline{\tau_0} \pm \Delta\tau_0 = 0.563 \pm 0.388\ (s)$$

- Direct measurement:

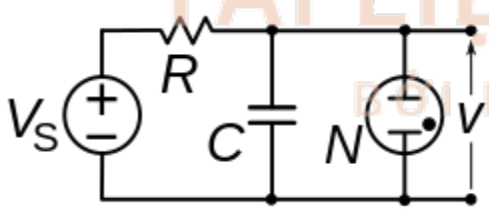
$$\tau_0 = \overline{\tau_0} \pm \Delta\tau_0 = 0.777 \pm 0.003\ (s)$$

$$C_x = \overline{C_x} \pm \Delta C_x = 0.566 \pm 0.008\ (\mu F)$$

$$R_x = \overline{R_x} \pm \Delta R_x = 3.132 \pm 0.066\ (M\Omega)$$

VI. Questions

1) <u>Describe the accumulate-released oscillation circuit using the neon light. Draw the circuit diagram and proven the equation to determine the period the accumulate-released oscillation.</u>



- The accumulate-released oscillator circuit is a simple electric oscillator circuit that generates a sawtooth output waveform. The circuit comprises:

+ A neon bulb that consists of a glass bulb seperated by neon at low pressure (10mmHg). It also has two metal electrodes A and K at a distance of 2 ÷ 3 mm apart from each other.

+ A protective resistor R with value reaching megaohm (MΩ) in series with neon light Ne

+ A capacitor (capacitance C) with microfara size (μF) parallel with neon light Ne

+ A constant electricity supply with an electric potential U_n.

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- Proof of the equation for finding the period of oscillation:

+ We have these basic formulas:

$$i = \frac{U_n - U}{R} \ (1),\ \ dq = i.dt \ (2), dq = C.dU \ (3)$$

+ Substituting (3) and (1) into (2), we have:

$$dt = R.C.\frac{dU}{U_n - U} \ (4)$$

+ Integrating (4) with t from timeframe t_1' to t_2 with electric potential values U_e and U_b between two poles of the capacitor C:

$$t \approx \int_{t_1'}^{t_2} dt = \int_{U_e}^{U_b} R.C.\frac{dU}{U_n - U} = \mathrm{R.C.}\ln\frac{\mathrm{U_n - U_e}}{\mathrm{U_n - U_b}}$$

2) <u>Present how to measure the "on" electric potential U_S and the "off" potential U_T of the neon light. Why need to connect the resistor in series to the neon light?</u>

- Firstly, we need to construct to accumulate-released oscillator circuit. We connect a resistor R = 100kΩ in series with the neon light and a voltmeter parallel to the neon light.

- Then we turn on the switch of the electric supply U_n to increase electric potential U by spinning the knob until the neon light glows. U is now equal to the on electric potential U_b of the neon light. We can measure this value.

- Finally we turn back the knob of the supply U_n to decrease the potential U between the two poles of neon light Ne until the light go out. U is now equal to the off electric potential U_e of the neon light. We can measure this value.

- A resistor is connected in series with the neon light in order to limit current through it when the light turns on and the capacitor discharges to prevent damage. If the resistor is connected parallel to the neon light, when the capacitor discharges, the resistor wouldn't be able to protect the circuit properly as the current only flows through the short circuit.

3) <u>Clarify the method of testing equation to determine the period of the accumulate-released oscillation circuit RC.</u>

4) <u>Present methods for determining the resistance and capacitance by the accumulate-released oscillation circuit using the neon light.</u>

- To determine unknown resistance R_x

+ Switch the sample resistor R_0 with R_x and turn on the switch K. Turn the knob of the supply U_n until voltmeter V reaches U_n = 90V and keep this value constant during the experiment and write down the period of oscillation. Press the knob "RESET" on the chronometer MC-963A. Then, chronometer MC - 963A automatically measure the interval time t_x of n = 50 of the period τ_x. Then finding the period τ_x of the accumulate-released oscillation of the circuit R_xC_0, measured by:

$$\tau_x = \frac{t_x}{n} = \frac{t_x}{50}$$

+ Then we can find

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$$R_x = R_0.\frac{\tau_x}{\tau_0} = R_0.\frac{t_x}{t_0}$$

- To determine unknown capacitance C_x

+ Switch the sample capacitor C_0 with C_x and turn on the switch K. Turn the knob of the supply U_n until voltmeter V reaches U_n = 90V and keep this value constant during the experiment and write down the period of oscillation. Press the knob "RESET" on the chronometer MC-963A. Then, chronometer MC - 963A automatically measure the interval time t_x of n = 50 of the period τ_x. Then finding the period τ_x of the accumulate-released oscillation of the circuit R_0C_x, measured by:

$$\tau_x = \frac{t_x}{n} = \frac{t_x}{50}$$

+ Then we can find

$$C_x = C_0.\frac{\tau_x}{\tau_0} = R_0.\frac{t_x}{t_0}$$

5) <u>Why not use this method to measure the small resistance and capacitance?</u>

- If R is too small, when the neon light is active, the current flowing through R will be large enough to keep it conducting without current from the capacitor, and the voltage across the neon light will never reach U_e so the neon light will never turn off which means the cycle can not continue.

- If C is too small, the capacitor will charge and discharge very quickly and thus make our measurement prone to inaccuracies and errors.

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